

Deployment of an Autonomous Underwater Vehicle Designed to Quantify Ocean Pollution

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Abstract—This paper aims to present the design and goals of an autonomous underwater vehicle for the final project of Harvey Mudd College’s Experimental Engineering course. This vehicle aimed to measure pollution by quantifying three different variables: pH, water turbidity, and water temperature. These values were obtained through an IR light emitter and photodiode turbidity sensor system, a linear temperature sensor, and a pH probe. The vehicle was deployed at Bernard Field Station and Scripps College Pool as test runs, and then data was collected at Dana Point Baby Beach to assess the effects of human activity on ocean water pollution. The robot’s autonomous pathway allowed it to dive down to a maximum depth of 2 meters, stopping at 2 waypoints at 0.5 meters and 1 meter to collect incremental sets of data. The autonomous pathway utilized proportional (P) control and a pressure sensor to monitor and correct the robot’s depth as it descended. Prior to deployment, calibration curves were made for each sensor in laboratory settings. At the deployment location, voltage values were collected from each sensor and then the calibration curves were used to determine the temperature, pressure, turbidity, and pH values. While it is possible to draw some conclusions from the sensor measurements, more trials and further sensor tuning would be needed to determine a more concrete conclusion. Based on the available sets of data, it can be determined that there indeed is some level of non-negligible human pollution at Dana Point, based on the discrepancies between the expected “healthy” values of pH, temperature, and turbidity of ocean water and the measured values collected with the sensors.

I. INTRODUCTION

The goal of this autonomous vehicle is to quantify the levels of human pollution at Dana Point. Human activity mainly impacts pollution levels by increasing the amount of carbon emissions in the atmosphere. In order to maintain equilibrium, the ocean begins absorbing the carbon dioxide as well, causing ocean acidification through the formation of carbonic acid from carbon dioxide and water.



The expected current pH of ocean water is 8.1, and at the current levels of carbon dioxide emissions, is expected to drop to 7.7 by 2100 [1]. Species that rely on basic aquatic environments to maintain their carbonate shells require an ideal pH of 8.1-8.3 [2].

The pH sensor is a silver chloride sensor that outputs a voltage in the millivolt range based on the pH levels of the solution it was immersed in. The probe measures the electric

potential across the internal glass electrode and the sample, which is then proportional to the overall pH of the solution. [3]

Increasing temperature changes the kinetic energy of the solution, which effects the number of hydronium ions that the pH sensor senses. [4] As a result, an accurate voltage to pH conversion requires a concurrent temperature reading. To accurately calibrate the pH sensor, a temperature sensor was necessary to relate pH to temperature. Since the expected time constant of the pH sensor is 40-60 seconds, a linear temperature sensor was chosen to create a simpler circuit, compared to a thermocouple or thermistor sensor circuit. A MCP9701A linear temperature sensor was chosen to collect temperature readings. The expected values of ocean temperature at Dana Point at this time of the year are around 18 °C [5]. pH is expected to change with temperature values due to higher temperatures increasing molecular activity and water dissociation into hydroxide and hydronium ions, contributing to lower pH as temperature rises. [3]

Turbidity of the ocean is typically used to quantify the cloudiness of water from suspended solids. In coastal areas such as Dana Point, higher turbidity values typically correspond to higher levels of runoff and pollution in the water. Typically, coastal water has turbidity values around 2 NTU, with significant runoff events resulting in 50-200 NTU levels of turbidity [6]. The turbidity sensor utilizes a pulsing IR1503 LED controlled by a 555 timer to provide a distinct light source that can be isolated from surrounding incident light. The light is allowed to pass through a column of seawater, and the transmitted light is then measured by 2 OP950 IR photodiodes positioned at 90° and 180° relative to the IR LED.

A MPX5700 pressure sensor was utilized to measure the depth of the robot in real time and provide a negative feedback control loop through a P control system. Trial and error was used to evaluate the ideal Kp value to minimize steady state error while avoiding excessive oscillation around the desired depth waypoint.

II. METHODS

A. Engineering Goal

The goal of this project was to quantify ocean pollution at different ocean depth levels at Dana Point. To quantify the physical and chemical effects of pollution on ocean water,

temperature, pH, and turbidity were all measured concurrently at selected waypoints.

B. Sensor Selection

The autonomous vehicle was outfitted with 4 distinct sensors to collect the relevant batches of data. These sensors were a pressure sensor, turbidity sensor, a pH sensor, and a temperature sensor.

All sensor circuits except the pH sensor were mounted onto a single protoboard that allowed for easier circuit assembly and organization. All sensor inputs were read by the analog pins of the Teensy 4.0. This limited the input voltage range to 0V to 3.3V, so all operational amplifiers were made with the intention of adjusting the output voltage range of the sensors to maximize the Teensy input range without going outside of it. The pH sensor circuit was made separately on a miniature breadboard to allow it to be enclosed in a Faraday cage.

The MPX5700 sensor was chosen to measure the depth of the robot and enable the robot to perform P control for desired depth values. The sensor measures depth by outputting a voltage that is linearly proportional to the pressure experienced by the sensor. This voltage reading is then converted into pressure, and then to the corresponding depth in water that would be required to induce that value of pressure.

The turbidity sensor integrated an IR1503 LED and 2 OP950 photodiodes. The photodiodes were placed at 90° and 180° relative to the LED to measure the amount of light scattered by the ocean water. This sensor was used to quantify the amount of sediment and particulates of ocean water as a way to measure the physical impact of ocean pollution.

The pH sensor was chosen to quantify acidification of ocean water from pollution. The pH sensor was integrated using a gain operational amplifier to amplify the raw millivolt signals to larger ranges that can be read by the Teensy. Due to budget constraints, this sensor had a tolerance of ± 0.1 pH value, which is significantly larger than any pH difference that will be experienced by the robot within a 2 meter depth range.

The temperature sensor was chosen to be the MCP9701A linear temperature for easy circuit integration. Temperature measurements were originally integrated in order to provide a correction factor for the pH sensor. However, post-experimental research showed that the correction for pH relative to temperature was negligible compared to the innate tolerances of the pH sensor itself. Therefore, the purpose of the temperature sensor pivoted to instead provide information on whether or not the ocean water was warmer relative to expected values, indicating rising global temperatures as a result of carbon dioxide emissions and other pollutants in the environment.

C. Pressure Sensor

The MPX5700 pressure sensor was incorporated in order to provide a measurement of the actual depth of the robot. Ultimately, this value was incorporated into a P control algorithm that allowed the robot to account for error between the actual depth and the desired depth.

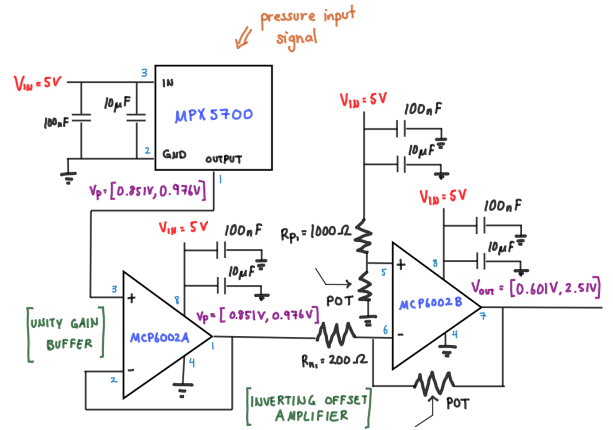


Fig. 1. Circuit schematic of pressure sensor.

The pressure sensor outputted a voltage that was linearly related to the pressure experienced by the sensor. Calibration was only done for a depth up to 1 meter due to lab constraints. The equation obtained was then extrapolated to the full 2 meter depth, which was the robot's planned maximum depth for deployment. During calibration, for a depth range of 0 to 1 meters, the pressure sensor outputted a raw voltage range of 0.85V to 0.97V. Using a gain factor of 26.16 and an offset of 0.74V, the output voltage was converted into a range from 1.89V to 2.51V. This was then extrapolated to incorporate the expected 2 meter depth, resulting in a final output voltage range of 0.60V to 2.51V, as shown in Fig. 1.

D. Turbidity Sensor

The turbidity sensor consisted of 3 main electrical components: 2 OP950 photodiodes, and 1 IR1530 LED. The overall sensor worked by placing the photodiodes at 90° and 180° with respect to the LED, and then measuring the amount of incident light on each photodiode as water passes between the LED and the photodiodes. Turbidity of the solution increases with higher amounts of sediment or "cloudiness" in the solution. As turbidity increases, the values measured by the 90° photodiode is expected to increase, as more sediment in the solution diffracts light away from the main trajectory of light.

A 555 timer set at 590 Hz with a 75% duty cycle was used to control the IR1503 LED with a square wave. This allowed the pulsing light from the LED to be distinguished from environmental light to provide a more accurate turbidity reading from the photodiodes. Each photodiode was amplified with a trans-impedance amplifier circuit with 47 kΩ resistor.

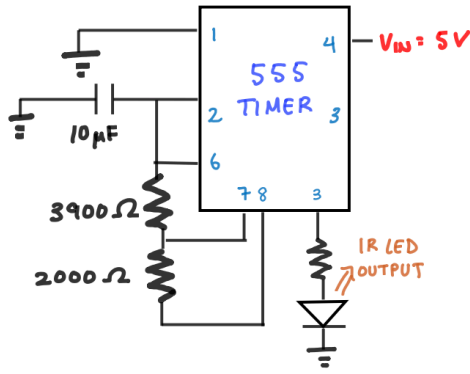


Fig. 2. 555 timer circuit with 590 Hz frequency and 75% duty cycle.

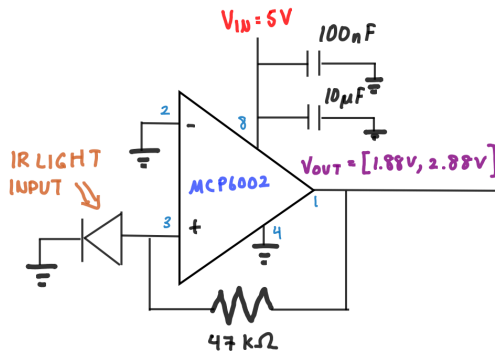


Fig. 3. Photodiode circuit that uses an inverting operational amplifier to amplify the signal from the photodiodes. An identical circuit was made for each photodiode, and both were integrated onto an MCP6002 chip.

Fig. 2 shows the 555 timer circuit, which was made on a TI NA555 timer. Fig. 3 illustrates the identical circuits made to amplify the signal from the OP950 photodiodes. The output voltage of the signals for turbidity ranges from 0 to 100 NTU are 1.88V to 2.88V, which is not ideal considering the full range of the Teensy is 0V to 3.3V. However, in order to amplify the signal further, a higher resistor value would be needed, which would raise the impedance of the overall circuit and ultimately result in greater current leakage. As a result, the decision was made to sacrifice signal voltage range and some level of precision for signal consistency.

$$TB = 1943 \times V \quad (2)$$

The turbidity calibration curve was obtained by using a mixture of milk powder and water in lab to create solutions with a range of turbidity values from 0 to 100 NTU. The equation (2) shows a linear relationship with an $R^2 = 0.926$. These solutions were measured using a turbidity meter as a ground truth measurement tool. The resulting calibration curve was then used at Dana Point to convert voltage values to turbidity NTU units (2).

E. pH Sensor

The pH sensor was purchased off Amazon due to budget constraints. Since the pH sensor outputs voltage on the range

of millivolts for each full pH point difference, a voltage reference set to 1.024V was used to artificially inflate the signal to a higher range. A REF35120QDBVR 1.024V voltage reference was chosen because of its low tolerances and overall stability. Adding the voltage reference to the initial pH voltage output allowed for the operational amplifier to have to add less of an offset, reducing the resistor values and impedance on the circuit. The expected raw sensor output voltage for a pH range of 6 to 9 after adding the voltage reference was 2.5V to 2.86V. To sufficiently amplify this signal, an inverting offset amplifier was used.

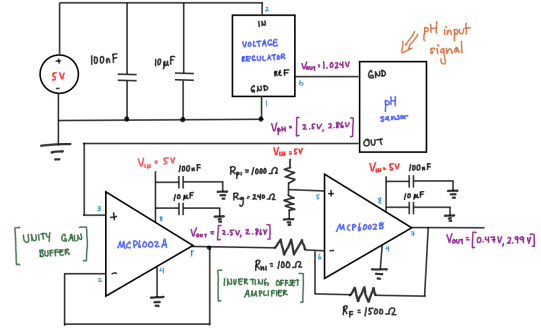


Fig. 4. Circuit schematic for the pH sensor.

As seen in Fig. 4, a unity gain buffer was used to reduce impedance from the pH sensor. Then, the inverting offset amplifier amplified the raw signal to a range of 0.47V to 2.99V, which falls within range of the Teensy voltage input range.

$$pH = 1.23 \times V + 6.37 \quad (3)$$

The calibration for the pH sensor was done with a set of buffer solutions with pH values 4, 7, and 10. Solutions were combined to achieve pH values between 6 and 9 to create the calibration curve shown in (3), which resulted in a linear relationship with $R^2 = 0.94$.

The pH sensor circuit had to be enclosed inside of a Faraday cage, due to the sensitive nature of the sensor. Since the pH sensor is outputting extremely small voltage, any outside noise could contribute to large error bounds. The Faraday cage was constructed of 3 layers of aluminum tape, which was then covered by 3 more layers of copper tape. The aluminum tape served the solid overall structure of the cage, while the copper tape provided added conductance and allowed for a better overall electrical insulation. The cage was also connected to ground to allow for a sink for external signal to flow away from the circuit. The circuit was created on a breadboard and then placed inside the cage, with only power, ground, signal in, and signal out wires exiting the Faraday cage.

F. Temperature Sensor

A linear temperature sensor was chosen for this circuit because of the pH sensor's large time constant. In a previous lab, the time constant for the MCP9701A linear temperature sensor was determined to be around 40-60 seconds [7]. Since the pH sensor is expected to also require a similar time

constant to acclimate to the environment, the linear temperature sensor was chosen over a thermocouple or thermistor for easier circuit assembly.

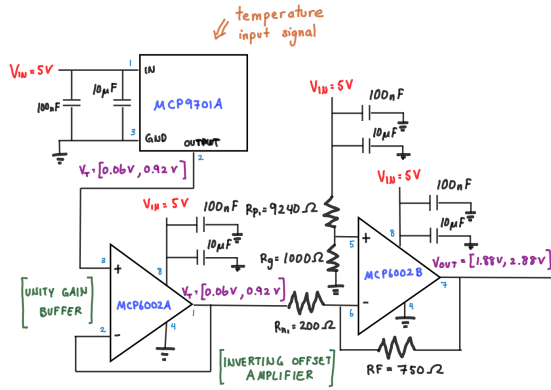


Fig. 5. Circuit schematic for temperature sensor.

The expected temperature range at Dana Point was 14°C to 24°C. As seen in Fig. 5, the temperature sensor's raw voltage range for this set of temperature values is 0.06V to 0.92V. A unity gain buffer was used to reduce impedance from the temperature sensor itself, and then used an inverting offset amplifier to amplify the signal to a 1.88V to 2.88V range.

$$T = -4.95 * V + 28.4 \quad (4)$$

The temperature calibration curve was done using freshwater baths in the laboratory setting, and the resulting linear relationship curve had $R^2 = 0.99$.

G. Mechanical Design

Since the main trajectory path of the robot was to dive downwards, the robot did not need a wide body for buoyancy. As a result, a simple box design was chosen that allowed the robot to remain upright while traversing up and down in water.

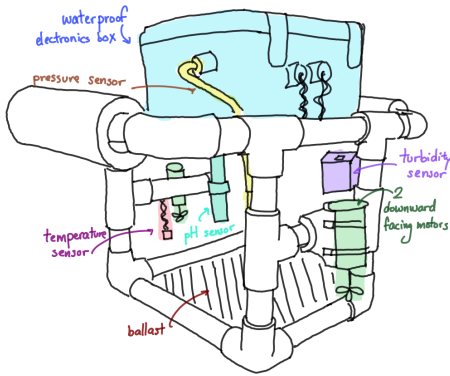


Fig. 6. Mechanical drawing for the robot design. Different components and sensors are not drawn to scale, but instead incorporated to give an overall image of what the robot looked like.

As seen in Fig. 6, the electrical component box was placed at the very top of the robot and secured with velcro and a mesh net. Since the box is the largest contributor to the robot's overall buoyancy, it was placed on top of the robot to prevent

rolling over in the water. Using a similar line of reasoning, the ballast for the robot was placed on a mesh net on the bottom of the robot in order to lower the center of mass and keep the robot upright. The box placed on top of the robot also enabled for easier access to the electronics for pre-deployment checks. Additional buoyancy was added by attaching pool noodles to the overall frame of the robot.

In incorporating and placing the sensors, the main objective was to keep each sensor as far away from the motors as possible. Running motors induce current noise on sensors because of the vibrations that occur inside the brushes as the motor rotates. This, combined with the inductance of the motor coils and leads, creates noise in nearby wires, including the wires that connect to the sensors on the robot [8]. However, due to the small shape of the robot, creating a large amount of distance between the sensors and motors proved to be very difficult. This problem was solved by including a software coded "motor off" period that allowed the sensors to collect data while the motors are off. Finally, the pH sensor was made to orient downwards to facilitate proper storage in buffer, as well as prevent the silver chloride solution inside from moving away from the glass probe.

H. Experimental Procedure

Deployments were performed in early morning times to prevent bright sunlight from interfering with turbidity readings with large amounts of incident lights. The deployment at Dana Point was performed at 8:30AM.

Prior to each deployment, the Teensy was connected to the computer via micro-USB cable, and the code was pushed onto the Teensy. Then, a dry test was performed, which involved keeping the Teensy plugged into the computer and reading off the serial output values. This allowed for quick troubleshooting to see which sensor was not connected or not reading proper values. To test each sensor, a series of manual tests were performed. To evaluate the pressure sensor, a team member blew on the pressure tube to artificially induce pressure and see if the sensor responds with an increased depth reading. The temperature sensor was pinched to observe a change in voltage reading. The pH sensor was immersed in a known buffer solution to ensure that it was reading the correct values. Finally, the turbidity sensor was blocked with a piece of plastic to see if the 180° photodiode showed a lower voltage reading.

Once the robot passed all the manual inspection procedures, the micro-USB was disconnected from the Teensy and then power was turned off. The robot was then brought to the deployment site, and then once the robot was ready for deployment, the motherboard was turned on. The software has a 30 second delay encoded into the start of the program, so it was important to quickly seal up the electronics box and then lower the robot into the water. Once the autonomous program started, the tether was monitored closely to ensure that the robot had enough slack to travel downward while maintaining a firm grip on the tether.

During deployment, the main goals were to monitor the marks on the tether that indicated the depth the robot was at. This served as the ground truth measurement of depth,

and served as a good check to see if P control was working as expected. The robot's expected trajectory was to reach a waypoint, hover for 40 seconds, turn off motors for 5 seconds, and then continue to the next waypoint. The final waypoint of the robot is set to 0 meters, so the robot is able to bring itself up to the surface before being recovered by a member of the deployment team. The robot was then quickly dried off, and the electronics box was opened to turn off the motherboard and remove the SD card to analyze the collected data.

III. RESULTS AND ANALYSIS

A. Results Overview

The overall results of the robot were collected from a single deployment at Dana Point. While a second deployment was attempted, the robot hit a part of the dock on the way down, breaking the seal of the electronics box and flooding all the electrical components. This deployment mishap was not noticed for at least 20 seconds, so by the time the robot was recovered, the electronics were already shorted for too long, and the robot could not be recovered.

B. Pressure Sensor Results

The initial calibration of the pressure sensor was done in laboratory settings at Harvey Mudd College, which is at an elevation level of about 1168 feet [9]. Dana Point has an elevation of around 144 feet, leading to about a 1000 foot difference in elevation and a significant difference in air pressure [10].

$$P = -0.848 \times V + 2.51 \quad (5)$$

Prior to the Dana Point deployment, the offset in the pressure calibration curve was adjusted in order to account for this change in neutral air pressure. The slope of the calibration curve is related to the density of the liquid, which changed from freshwater in lab to saltwater at Dana Point. However, since the density of saltwater compared to freshwater is only a 2% difference, it was considered to be negligible considering the much larger steady state error of the pressure sensor P control [11]. As a result, the calibration equation converting voltage to depth at Dana Point was only adjusted for the added neutral pressure from the elevation difference (5).

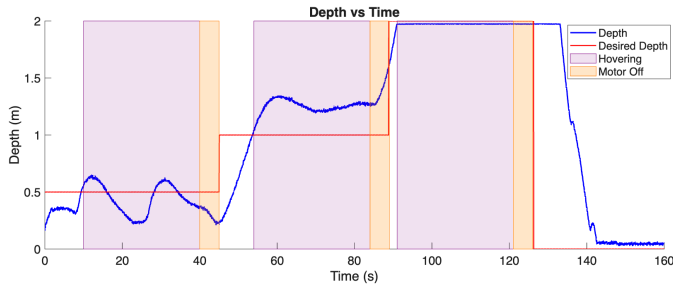


Fig. 7. Graph of desired depth versus measured depth. The purple regions indicate the hovering period that lasts for 30 seconds, which is designed to allow the sensors to acclimate to the environment due to the large time constants. The orange region is the “motor off” period coded to last for 5 seconds before moving to the next waypoint.

From Fig. 7, it is seen that the pressure sensor performed good P control and allowed the robot to stay within range of the desired depth. A steady state error is present, which is expected given the shortcomings of P control.

In one of the test deployments, the robot was taken to Scripps pool and compared the results of P control and PID control.

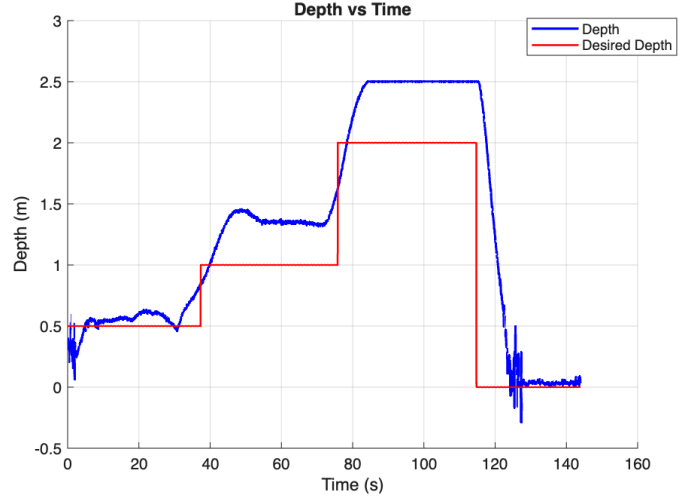


Fig. 8. P Control test performed at Scripps Pool.

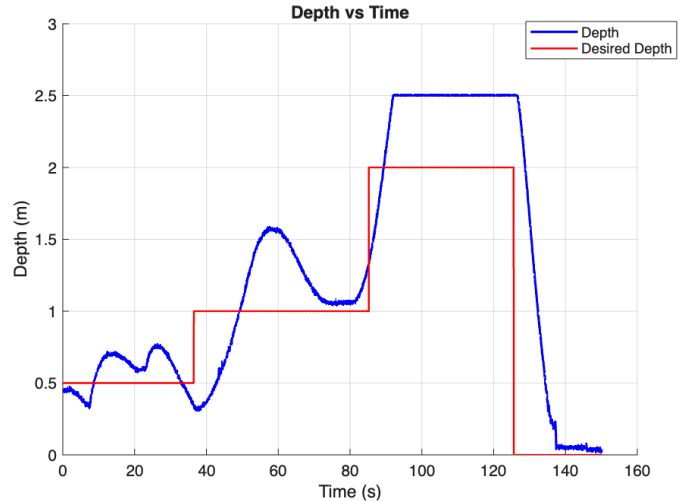


Fig. 9. PID control test performed at Scripps Pool.

From Fig. 8, it is clear that P control results in a consistent amount of steady state error. From Fig. 9, PID control clearly reduces the steady state error to a nearly negligible value. However, PID control also resulted in large oscillations. Given that the chosen sensors require a lot of time for environmental acclimation, P control was determined to be the better option because it allows the sensors to be in a similar environment for longer.

C. Turbidity Sensor Results

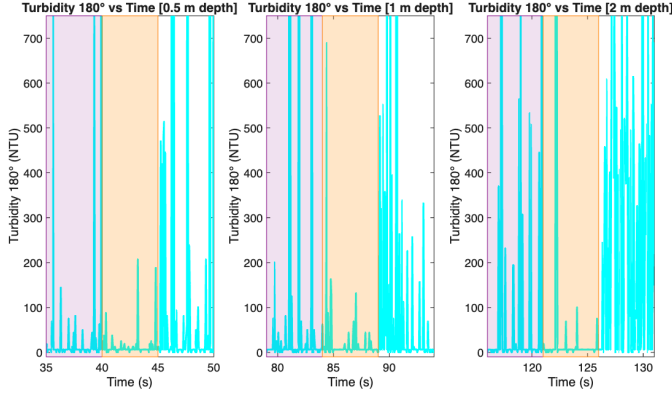


Fig. 10. Turbidity sensor readings at Dana Point. The motor off regions are selected and displayed here to provide a smaller x-axis range and provide clearer data analysis.

Turbidity measurements were difficult to obtain due to the challenges from waterproofing of the sensor. The collected data at each depth point from Fig. 10 was averaged to provide an overall image of the turbidity. As seen in Fig. 10, there were certain large spikes in the data. From the collected voltage readings, these spikes corresponded to 3.3V input readings, indicating that the sensor temporarily railed out. These values were attributed to waterproofing errors that caused temporary shorts in the circuits, so those values were removed from the

TABLE I
AVERAGE TURBIDITY SENSOR DATA AT EACH DEPTH WAYPOINT

Depth (m)	Turbidity (NTU)
0.5	12.72
1	21.97
2	110.83

From Table I, it is clear that as the robot traversed downwards, turbidity increased. This aligns with the initial expectations because if there were sediment in the water, then due to gravity, the expectation is for sediment concentration to increase as the robot traverses downward. A possible confounding factor could be that incident light is more prevalent at the surface of the water, compared to deeper depths. However, due to the use of the square-wave output from the IR1503 LED, and the corresponding oscillations seen in the turbidity sensor readings, it is unlikely that the sensors are reacting to environmental light, and more likely that the sensors are reacting to the intended IR1503 LED light source.

The ground truth readings of turbidity, obtained from team 24, indicated a surface turbidity value of around 23.25 NTU, which aligns relatively well with the measured value at 0.5 meters depth. In addition, as mentioned above, the trend of the turbidity value readings aligns with the expected values. However, it is noted that aliasing occurred in the collection of the turbidity data.

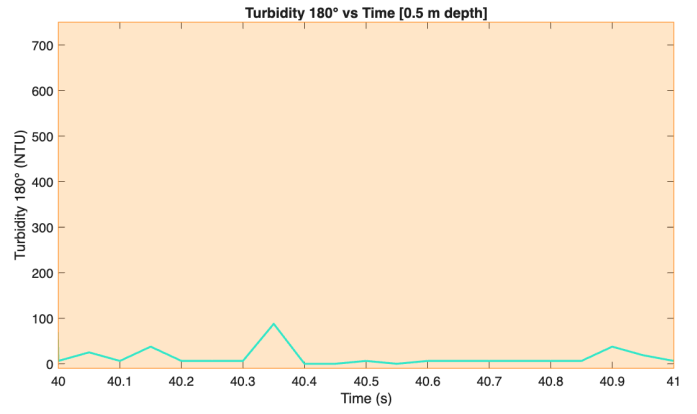


Fig. 11. Zoomed in turbidity data at 0.5 m waypoint from 40 seconds to 41 seconds.

As shown in Fig. 11, turbidity is expected to match the square wave shape of the input from the IR LED. However, it is clear here that the Teensy is not sampling quickly enough to capture the entire square wave, resulting in sharp triangle shaped peaks. Therefore, this method of aliasing could result in overall attenuation of the true values of the sensor readings.

D. Temperature Sensor Results

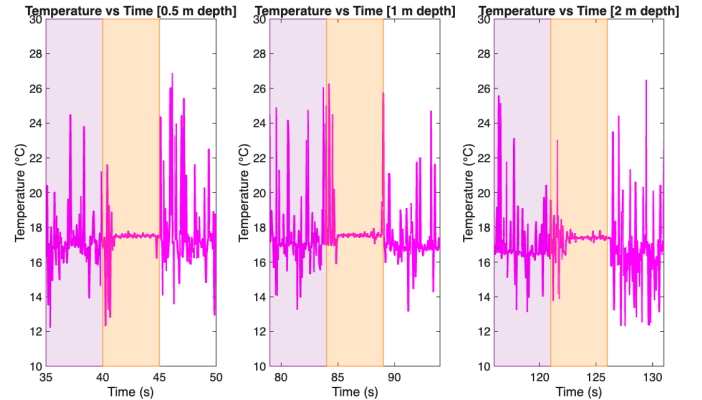


Fig. 12. Temperature sensor readings at Dana Point.

The temperature sensor readings in Fig. 12 show a significant reduction of noise in motor off periods. This is as expected, since linear temperature sensors must transmit low-voltage readings across large distances, acting as antennae for nearby current sources. Their linear relationship mean that any noise added to the signal is amplified into large error values [12]. In addition, the temperature sensor was relatively close to the motors, making it most susceptible to motor noise.

TABLE II
AVERAGE TEMPERATURE SENSOR DATA AT EACH DEPTH WAYPOINT

Depth (m)	Temperature (°C)
0.5	17.34
1	17.86
2	17.33

As seen in the Table II, temperature did not show any significant trend as the robot traversed downwards. This does not align with initial expectation of temperature decreasing as the robot traversed downwards. The collected data was compared to data collected by team 24, who obtained lowering temperature values as the robot traversed downwards. A possible explanation for the incorrect temperature readings could be the the waterproofing method of the sensor. The sensor was wrapped in a lot of Parafilm, which could act as a temperature insulator and increasing the temperature constant beyond the expected 40-60 second range. As a result, since the increased time constant was not accounted for, the temperature readings at each waypoint were not accurate.

E. pH Sensor Results

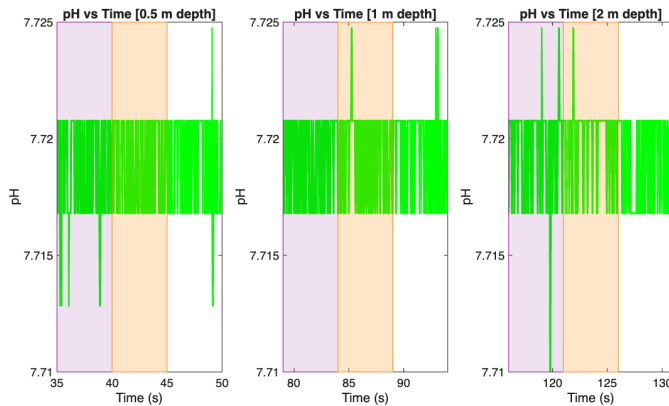


Fig. 13. pH sensor readings at Dana Point.

The pH sensor showed a significant amount of noise in Fig. 13, even in the orange motor off regions. This is likely due to the overall sensitivity of the pH sensor. Even with the implementation of a Faraday cage, the sensor was still relatively unreliable. The data could be averaged to obtain an overall value of 7.2 for all 3 depth waypoints. The ground truth pH value was stated to be between 7.9 and 8.0.

After deployment, a new reference document was obtained that explained the shortcomings of the pH sensor in ocean water. The pH sensor measures pH by measuring the electric potential across the reference electrode and the sample solution it is immersed in. pH is determined by the concentration of H^+ ions, and in most solutions, this is the most abundant positively charged cation. However, in ocean water, the high ionic concentration of sodium ions adds another source of cations in solution. The United States Geological Survey estimates around 35g of sodium chloride per liter of water, which amounts to a weight concentration of around 3.5% [13]. This non-negligible concentration of ions in the water causes the pH sensor to be unable to discriminate between the sodium ions and the hydrogen ions, resulting in artificially lower pH values being read. It was recommended to account for this discrepancy by adding a 0.5 offset to collected pH values. This places the corrected values around 7.7, as shown in Fig. 13.

In addition to ionic strength, the pH values were also affected by the temperature of the solution. Tests performed in

lab after deployment showed that at a temperature of around 14°C , a voltage sensitivity was expected to have a slope of around 57 mV/pH [4]. However, since no concurrent temperature readings were obtained, this source difference was not able to be accounted for.

pH values remained consistent at all depth values, which makes sense since pH is not expected to change much in a small range of depth. However, this value is significantly lower than expected, indicating some level of ocean acidification occurring at the Dana Point Dock.

IV. CONCLUSION

A. Overall Results

While many aspects of the data collection were not ideal, there were still meaningful conclusions to be drawn from the collected data. Ideally, multiple trials would be able to be collected and compared to provide the most accurate set of data.

Among all the sensors, the pressure sensor was found to be the most accurate. The pressure sensor was an integral part of the diving P control system, so being able to dive accurately to the expected waypoints is critical for a successful deployment.

While the temperature sensor and the pH sensor both recorded data as expected, both values were slightly off from the expected ground truth values. As mentioned before, temperature sensor's inaccuracy was believed to be due to the waterproofing system used that inadvertently increased the time constant by insulating the sensor. Retroactively, it was noticed that the pH sensor was calibrated incorrectly due to a set of expired pH strips. Due to the flooding of the electronics box, the sensor was unable to be recalibrated after deployment. However, with proper calibration, it is expected that the pH sensor would likely read accurate values.

The most unsuccessful sensor was likely the turbidity sensor. From the data, it is clear that the Teensy sampling rate caused some level of aliasing in the data. In addition, there are large peaks in the data that reach 3.3V, indicating that the sensor must be railing out. During the build process, there was a lot of difficulty successfully waterproofing the turbidity sensor, so the spikes in the data are hypothesized to be a result of the sensor shorting out due to water ingress into the sensor.

B. Significance of Results

From the collected data, it can be concluded that there is some non-negligible level of water pollution in the water at Dana Point. Even simply comparing the ground truth values to the expected ranges at this time of year, there was definitely increased water temperature, increased turbidity, and lowered pH values. Since increased carbon emissions is likely the source of water pollution, it makes sense that there is also overall warming of the ocean water. In addition, the lower pH value indicates that there is some level of ocean acidification. Finally, increased turbidity indicates a higher level of sediment in the water, which is likely a result of increased pollution in the water.

C. Recommendations for Future Work

Recommendations for future work include expanding these sensors out to become more permanent fixtures in the ocean. These sensors can be incorporated into sensor stations at various popular beach and tourist locations. This could be beneficial by allowing public access towards real time data about water pollution. Ultimately, building these sensor stations would increase equitable data distribution, and allow people to make more informed decisions about their health.

D. Acknowledgements

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